A Course End Project Report on

Personal Expense Tracker

A9602 - Object Oriented Programming through Java Laboratory

Submitted by

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Managing personal expenses manually can be time-consuming and inefficient, making it difficult to track daily spending and maintain a proper budget. This project presents a simple and effective solution for managing personal finances through a Personal Expense Tracker. The proposed system enables users to record their daily expenses, categorize spending, and calculate total expenses using a user-friendly application developed with Python. The system helps users maintain expense records, identify spending patterns, and monitor their financial activities. It reduces the effort required for manual calculations and improves accuracy in expense management. This project aims to provide an accessible, cost-effective, and efficient solution for students, employees, and individuals who want to manage their money effectively. By organizing expenses into categories such as food, transportation, education, shopping, and entertainment, the system helps users understand where their money is being spent. Additionally, this project explores potential future enhancements, such as database integration, graphical reports, monthly budget alerts, and mobile application support, to improve its functionality and usability. The successful implementation of this system contributes to better financial awareness, improved budgeting habits, and responsible money management.

Keywords: Personal Expense Tracker, Expense Management, Python, Budget Planning, Financial Management, Expense Categorization, Database Integration.

1 Introduction

Managing personal expenses is an important part of daily life. However, tracking daily spending manually can be time-consuming and may lead to errors in calculations. This project addresses these problems by developing a Personal Expense Tracker using Java. The system allows users to record their daily expenses, categorize spending, and calculate total expenses easily. It helps users monitor their financial activities and understand their spending habits. The proposed system provides a simple, affordable, and efficient solution for students, employees, and individuals who want to manage their money effectively.

2 Literature Review

Several existing methods are used for personal expense management, including paper-based records, spreadsheets, and mobile expense-tracking applications. Traditional methods require manual calculations and may make it difficult to maintain and analyze expense records. Spreadsheet-based systems provide better organization but often require users to enter formulas and manage data manually. Modern expense-tracking applications provide automated calculations, expense categorization, and graphical reports. However, some applications include unnecessary features or require paid subscriptions. A Java-based Personal Expense Tracker provides a simple and cost-effective alternative by allowing users to record expenses, view spending details, and calculate total expenditure. Future integration with database systems can further improve data storage and retrieval.

3 Objectives

- To develop a simple and user-friendly Personal Expense Tracker using Java.
- To record daily expenses with details such as amount, category, date, and description.
- To calculate the total expenses automatically.
- To categorize expenses such as food, transportation, education, shopping, and entertainment.
- To help users monitor their spending habits and manage their budgets effectively.
- To explore future enhancements such as database integration, graphical reports, and budget alerts.

4 Methodology

4.1 Block Diagram

The system consists of:

- **Java Application:** Processes expense details and manages the application logic.
- **User Interface:** Allows users to enter, view, and manage their daily expenses.
- **Data Storage:** Stores expense records using files or a database such as MySQL.
- **Calculation Module:** Calculates total expenses and category-wise spending.

4.2 Circuit Diagram and Components

4.2.1 Components Used

- Java Application
- User Interface
- Data Storage
- Calculation Module

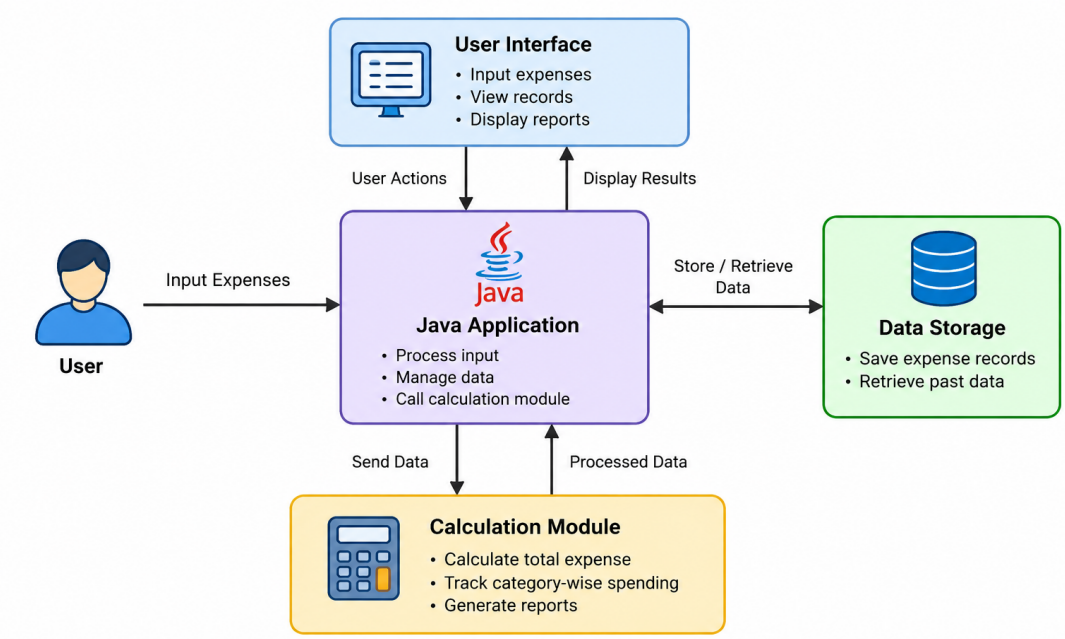


Figure 1: Personal Expense Tracker Circuit Diagram

Figure 1: Personal Expense Tracker Circuit Diagram

Figure 1 illustrates the Personal Expense Tracker system developed using Java. The system enables users to enter their daily expenses, including the date, amount, category, and description. The Java application processes the entered information, calculates total expenses, and organizes spending records by category. The expense details are stored in a text file or MySQL database for future reference. The system generates daily and monthly expense summaries to help users monitor their spending habits and manage their budgets effectively. This application is useful for students, employees, and individuals who want to maintain their personal financial records efficiently.

4.3 Working Principle

1. The user opens the Personal Expense Tracker application developed using Java.
2. The user enters expense details such as date, amount, category, and description.
3. The Java application processes the entered data and calculates the total expenses.
4. The expense records are stored in a text file or MySQL database for future reference.
5. The system displays daily and monthly expense summaries to help users monitor their spending and manage their budget.

4.4 Software Implementation

4.5 Java Program

```
import java.io.*;
import java.time.LocalDate;
import java.time.format.DateTimeParseException;
import java.util.*;
public class PersonalExpenseTracker {
    static Scanner sc = new Scanner(System.in);
    static ArrayList<Expense> expenses = new ArrayList<>();
    static final String FILE_NAME = "expenses.txt";
    static class Expense {
        int id;
        String date;
        String category;
        String description;
        double amount;
        Expense(int id, String date, String category,
                String description, double amount) {
            this.id = id;
            this.date = date;
            this.category = category;
            this.description = description;
            this.amount = amount;
        }
    }
    public static void main(String[] args) {
        loadExpenses();
        while (true) {
            System.out.println("\n===== PERSONAL EXPENSE TRACKER =====");
            System.out.println("1. Add Expense");
            System.out.println("2. View All Expenses");
            System.out.println("3. View Total Expenses");
            System.out.println("4. Category-wise Summary");
            System.out.println("5. Delete Expense");
            System.out.println("6. Exit");
            System.out.print("Enter your choice: ");

            int choice = readInt();

            switch (choice) {
                case 1:
                    addExpense();
                    break;
                case 2:
                    viewExpenses();
                    break;
                case 3:
                    showTotal();
                    break;
                case 4:
```

```
                categorySummary();
                break;
            case 5:
                deleteExpense();
                break;
            case 6:
                saveExpenses();
                System.out.println("Expenses saved. Thank you!");
                return;
            default:
                System.out.println("Invalid choice. Try again.");
        }
    }
}
static int readInt() {
    while (true) {
        try {
            return Integer.parseInt(sc.nextLine().trim());
        } catch (NumberFormatException e) {
            System.out.print("Enter a valid whole number: ");
        }
    }
}

static double readAmount() {
    while (true) {
        try {
            double amount =
                Double.parseDouble(sc.nextLine().trim());

            if (Double.isFinite(amount) && amount > 0) {
                return amount;
            }
        } catch (NumberFormatException e) {
            // Ask again if input is not a number.
        }

        System.out.print("Enter a valid amount greater than zero: ");
    }
}

static void addExpense() {
    System.out.println("\n--- Add New Expense ---");

    System.out.print("Enter date (YYYY-MM-DD): ");
    String date = sc.nextLine().trim();

    try {
        LocalDate.parse(date);
    } catch (DateTimeParseException e) {
        System.out.println("Invalid date format.");
```

```
            return;
        }

        System.out.println("Categories: Food, Transport, Education, Shopping, Ot
        System.out.print("Enter category: ");
        String category = sc.nextLine().trim();

        if (category.isEmpty()) {
            System.out.println("Category cannot be empty.");
            return;
        }

        System.out.print("Enter description: ");
        String description = sc.nextLine().trim();

        System.out.print("Enter amount (Rs.): ");
        double amount = readAmount();

        int id = 1;
        for (Expense e : expenses) {
            if (e.id >= id) {
                id = e.id + 1;
            }
        }

        // Avoid the file delimiter in saved text.
        category = category.replace("|", "/");
        description = description.replace("|", "/");

        expenses.add(new Expense(
            id, date, category, description, amount
        ));

        saveExpenses();
        System.out.println("Expense added successfully!");
    }

    static void viewExpenses() {
        System.out.println("\n--- All Expenses ---");

        if (expenses.isEmpty()) {
            System.out.println("No expenses recorded.");
            return;
        }

        System.out.printf("%-5s %-12s %-15s %-22s %10s%n",
                "ID", "Date", "Category", "Description", "Amount");
        System.out.println("-----------------------------------------------------------------

        for (Expense e : expenses) {
            System.out.printf("%-5d %-12s %-15s %-22s %10.2f%n",
```

```
                e.id, e.date, e.category,
                e.description, e.amount);
        }
    }

    static void showTotal() {
        double total = 0;

        for (Expense e : expenses) {
            total += e.amount;
        }

        System.out.printf("Total Expenses: Rs. %.2f%n", total);
    }

    static void categorySummary() {
        if (expenses.isEmpty()) {
            System.out.println("No expenses recorded.");
            return;
        }

        Map<String, Double> summary = new TreeMap<>();

        for (Expense e : expenses) {
            summary.put(
                e.category,
                summary.getOrDefault(e.category, 0.0) + e.amount
            );
        }

        System.out.println("\n--- Category-wise Summary ---");

        for (Map.Entry<String, Double> entry : summary.entrySet()) {
            System.out.printf("%-20s Rs. %.2f%n",
                    entry.getKey(), entry.getValue());
        }
    }

    static void deleteExpense() {
        System.out.print("Enter the expense ID to delete: ");
        int id = readInt();

        for (int i = 0; i < expenses.size(); i++) {
            if (expenses.get(i).id == id) {
                expenses.remove(i);
                saveExpenses();
                System.out.println("Expense deleted successfully!");
                return;
            }
        }
```

```
        System.out.println("Expense ID not found.");
    }

    static void saveExpenses() {
        try (PrintWriter writer =
                 new PrintWriter(new FileWriter(FILE_NAME))) {

            for (Expense e : expenses) {
                writer.println(e.id + "|" + e.date + "|" +
                    e.category + "|" + e.description + "|" + e.amount);
            }

        } catch (IOException e) {
            System.out.println("Error saving expenses: " + e.getMessage());
        }
    }

    static void loadExpenses() {
        File file = new File(FILE_NAME);

        if (!file.exists()) {
            return;
        }

        try (BufferedReader reader =
                 new BufferedReader(new FileReader(file))) {

            String line;

            while ((line = reader.readLine()) != null) {
                String[] data = line.split("\\|", 5);

                if (data.length == 5) {
                    try {
                        expenses.add(new Expense(
                            Integer.parseInt(data[0]),
                            data[1],
                            data[2],
                            data[3],
                            Double.parseDouble(data[4])
                        ));
                    } catch (NumberFormatException e) {
                        System.out.println("Skipping an invalid saved record.");
                    }
                }
            }

        } catch (IOException e) {
            System.out.println("Error loading expenses: " + e.getMessage());
        }
    }
```

```
}
Output:
====== PERSONAL EXPENSE TRACKER ======
1. Add Expense
2. View All Expenses
3. View Total Expenses
4. Category-wise Summary
5. Delete Expense
6. Exit

Enter your choice: 1

--- Add New Expense ---
Enter date (YYYY-MM-DD): 2026-10-25
Categories: Food, Transport, Education, Shopping, Other
Enter category: Food
Enter description: Lunch
Enter amount (Rs.): 150

Expense added successfully!
```

5 Results and Discussion

The Personal Expense Tracker application efficiently records and processes expense details entered by the user. The system calculates total expenses and organizes records based on categories such as food, transportation, education, and shopping. It displays expense summaries to help users understand their spending patterns and manage their budgets effectively.

The system stores expense records in a text file or MySQL database, allowing users to access previously entered information. Its performance depends on the implementation, the amount of stored data, and the storage method used. The application can be enhanced by adding graphical charts, monthly budget limits, automatic expense categorization, and alerts when spending exceeds the specified budget. In the future, cloud database integration could enable secure access to expense records across multiple devices, making the system more convenient for students, employees, and other users.

6 Mapping POs and SDGs

6.1 Program Outcomes (POs) Mapping

Table 1: Mapping of Program Outcomes (POs) to Project Relevance

PO No.	Program Outcome	Relevance
PO1	Engineering Knowledge	Applied Java programming and computational concepts to develop the expense tracker.
PO2	Problem Analysis	Identified problems in manual expense tracking and budget management.
PO3	Design/Development of Solutions	Designed a Java application to record, categorize, and calculate personal expenses.
PO4	Investigation of Complex Problems	Tested expense calculations and verified the accuracy of generated summaries.
PO5	Modern Tool Usage	Utilized Java, an IDE, and file handling or a database for application development.
PO6	Engineer and Society	Supports users in monitoring spending and improving personal financial management.
PO8	Ethics	Encourages responsible handling and privacy of personal financial records.
PO10	Communication	Presents expense details and summaries in a clear, understandable format.
PO11	Project Management and Finance	Applies basic budgeting and financial management concepts.

6.2 Sustainable Development Goals (SDGs) Mapping

Table 2: Mapping of Sustainable Development Goals (SDGs) to Project Relevance

SDG No.	Goal	Relevance
SDG 4	Quality Education	Helps students learn digital financial management and budgeting skills.
SDG 8	Decent Work and Economic Growth	Encourages better financial planning and responsible management of personal expenses.
SDG 9	Industry, Innovation, and Infrastructure	Promotes the use of software technology to develop a digital expense management solution.
SDG 12	Responsible Consumption and Production	Encourages users to monitor spending and make more informed purchasing decisions.

7 Conclusion

This project successfully implements a Personal Expense Tracker using Java, simplifying the process of recording and managing daily expenses. The system is cost-effective, efficient, and user-friendly, allowing users to categorize expenses, calculate total spending, and view expense summaries. It helps users monitor their spending habits and improve personal budget management. Future enhancements can include graphical reports, monthly budget alerts, automatic expense categorization, and cloud database integration to improve the system's functionality, accessibility, and scalability.

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